

Observability, guardrails, evals, and the golden dataset

Revision 3 named the production tools. This revision specifies the four quality planes that make the Axe MCP + LangGraph agent operable: how you see a run, how you stop a bad run, how you score a change, and how the fixture set that scores it is built and versioned.

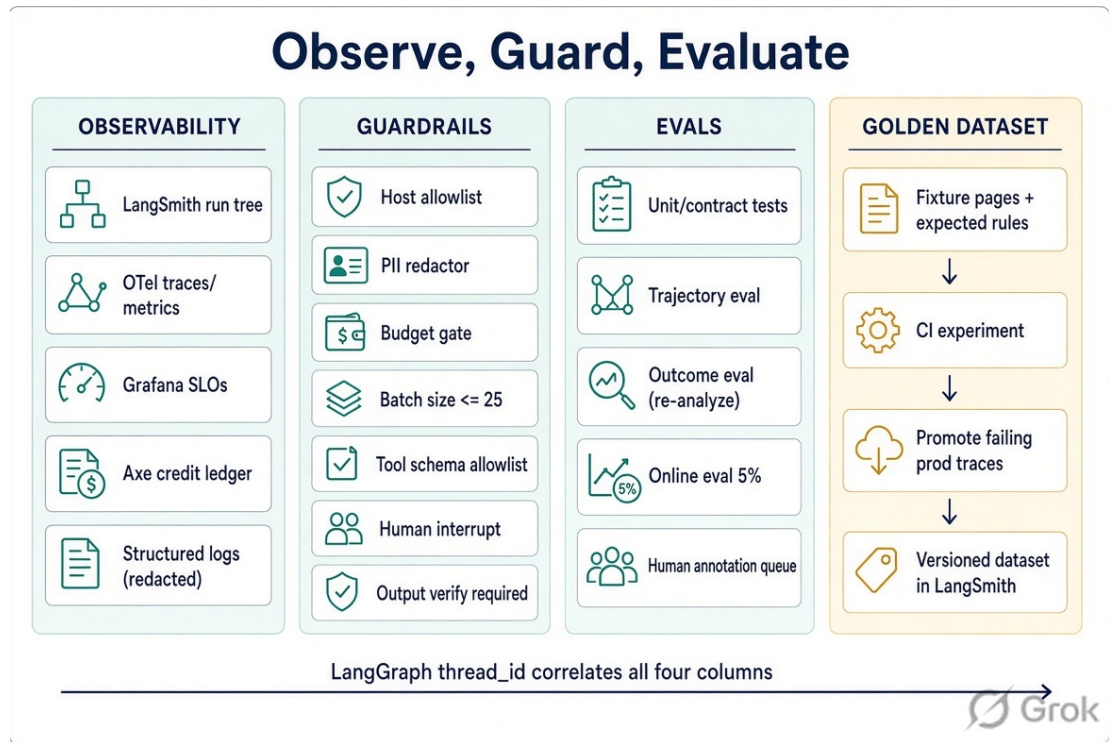


Figure 1. Four planes share one correlation key: LangGraph thread_id. Nothing in this brief is a separate product silo.

1. Correlation model

Every artifact — LangSmith span, OTel metric, Axe credit debit, screenshot object, GitHub Check, eval score — carries the same identifiers. If you cannot join them, you do not have observability; you have four dashboards.

Field	Example	Lives on
thread_id	acme/web#8421#/checkout#2026-09-05T08:11Z	graph state, LangSmith parent run, logs
repo / sha / pr	acme/web @ 9f3c... / PR 8421	CI, traces, Checks
url	https://pr-8421.preview.example/checkout	analyze args, dataset case
prompt_id / graph_sha / model	plan@v12 / graph@a91 / gpt-x-pinned	LangSmith metadata
credit_budget / credits_used	40 / 17	graph state, cost dashboard
dataset_version	golden-a11y@2026.09.1	CI experiment name

2. Observability plane

Split signals by the question they answer. LangSmith is the agent brain dump. OTel/Grafana is the machine. Axe Account Portal is the Deque meter. Structured logs are the audit tape. Do not send browser boot time only to LangSmith, and do not send prompt text only to Splunk.

Question	Signal	Backend
What did the agent decide?	Run tree: intake → analyze → cluster → remediate → apply → verify	LangSmith
Which prompt and model?	prompt_id, model pin, token counts on each LLM span	LangSmith
Did the tool call match the schema?	MCP args / results as child spans (redacted)	LangSmith
Is the worker farm healthy?	queue depth, browser boot p95, disk, CPU	OTel → Prometheus / Grafana
How long did analyze take?	span duration + OTel histogram axe.analyze.ms	both, same thread_id
What did Deque charge?	credits_used per remediate / IGT	graph state + Axe portal
What did the page look like?	screenshot object key, not the bitmap in the trace	object store + link in LangSmith
Who approved the patch?	human_decision on the interrupt span	LangSmith + checkpointer

Trace shape (normative)

```

LangSmith parent run
  name: a11y_thread
  extra: {thread_id, repo, sha, pr, url, prompt_id, graph_sha, model}
  ■■■ span intake
  ■■■ span analyze      tool=axe.analyze      extra.redacted=true
  ■■■ span cluster      extra.batch_sizes=[18]
  ■■■ span budget_gate  extra.credits_used=12 credits_budget=40
  ■■■ span remediate    tool=axe.remediate  extra.n=18
  ■■■ span human_gate   extra.decision=approve_subset
  ■■■ span apply        extra.files=[Checkout.tsx]
  ■■■ span verify       tool=axe.analyze      extra.remaining_serious=1
  ■■■ span report       extra.langsmith_url, screenshot_keys[], check_url

```

Figure 2. Required run tree. Missing verify span is itself an SLO failure.

What never goes in a trace

- before.fill.value, cookie values, Authorization headers, raw access tokens.
- Full page HTML. Keep selectors + a short snippet or a hash; store the rest in the checkpointer if a specialist needs it.
- Unredacted screenshots of authenticated views in a public LangSmith workspace. Use signed object-store URLs with TTL.

3. Guardrails plane

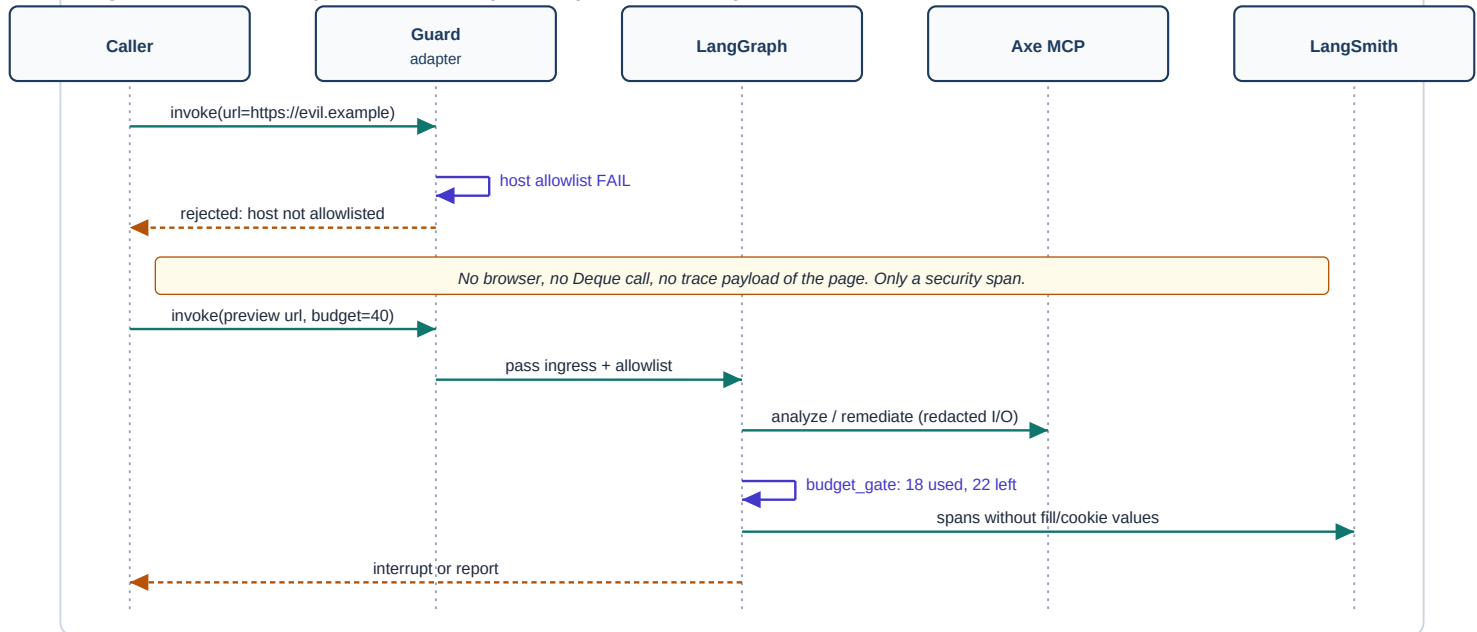
Guardrails are code on the graph edges, not instructions in a system prompt. A prompt can be ignored. A failed conditional edge cannot. Layer them so a single missed check is not the whole safety case.

Layer	Rail	Enforced at	On failure
Ingress	Caller authn/z; repo allowlist	API / Platform	403, no thread
Target	URL host allowlist (preview, localhost, listed staging)	intake node	reject; do not launch browser
Target	No production host unless explicitly flagged	intake node	reject
Privacy	PII / secret redactor on every tool I/O	MCP adapter	drop field; flag span
Tool	Only analyze, remediate, listed file/git tools	tool router	unknown tool = error
Tool	MCP JSON schema validation	LangChain StructuredTool	do not call Axe
Spend	credit_budget and max_rounds in state	budget_gate edge	report + interrupt
Spend	remediate batch size 1–25	cluster node	split or drop
Spend	Chromium concurrency semaphore	worker	queue; never stampede
Apply	IDE graph must interrupt before write	human_gate	wait for Command(resume)
Apply	CI graph auto-applies only safe_rules	compiled graph variant	leave the rest as findings
Close	verify after every apply	graph edge, not a prompt	thread marked incomplete

Layer	Rail	Enforced at	On failure
Close	cannot END with apply and no verify span	report node + eval	fail the Check

Two compiled graphs, one codebase: `ide_graph` has `human_gate` on; `ci_graph` has `auto_approve=safe_rules` and `fail_on=serious`. Do not toggle this with a sentence in the planner prompt.

Figure 3. Guardrails reject before the expensive path; traces stay redacted

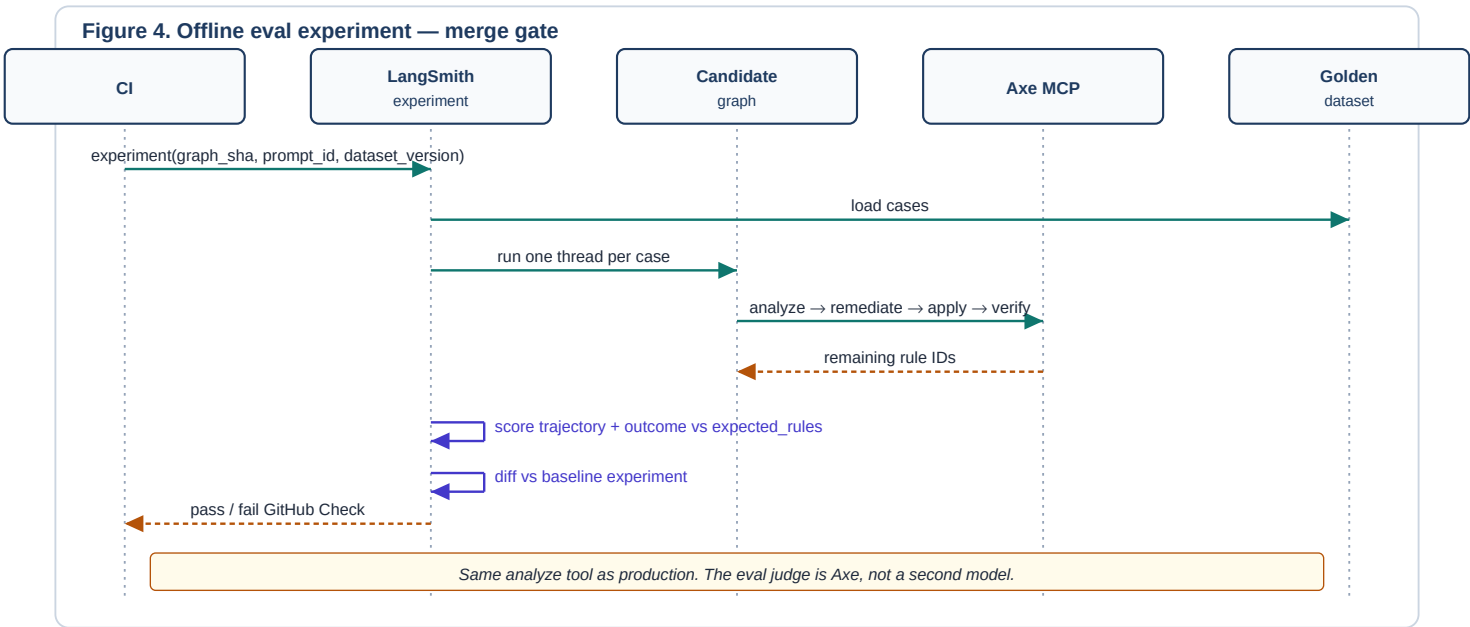


4. Evals plane

Agent evals are not “did the paragraph sound right.” This agent has a cheap, deterministic judge: run `analyze` again. LLM-as-judge is reserved for questions axe cannot answer (wrong file, patch style). Five layers, all required by the time a second team uses the graph.

Layer	What it scores	Judge	Gate
Unit / contract	MCP schemas; redactor; cluster batch ≤ 25 ; allowlist	pytest	every commit
Trajectory	analyze before remediate; verify after apply; no unknown tools; IGT only if flagged	LangSmith trajectory evaluator on the run tree	CI experiment
Outcome	serious+critical count dropped, or cap hit with coded reason	Axe analyze on the same URL (the golden expected_rules)	CI experiment
Cost	credits and tokens per fixed serious issue under budget	state fields + LangSmith token metrics	CI + weekly report
Online	same trajectory + cost checks on ~5% of live threads	LangSmith online evaluator	alert, not merge gate
Human	was this code_fix acceptable to a specialist?	annotation queue	dataset labels

Fail the PR if the candidate graph loses outcome score against the baseline experiment on the pinned `dataset_version`, or if trajectory completeness drops. Do not fail the PR on the online sample — that is drift detection, not a flaky gate.



5. Golden dataset

The dataset is a versioned product, not a folder of HTML files on someone's laptop. It is the spec the outcome evaluator scores against. If the dataset is wrong, every eval is theatre.

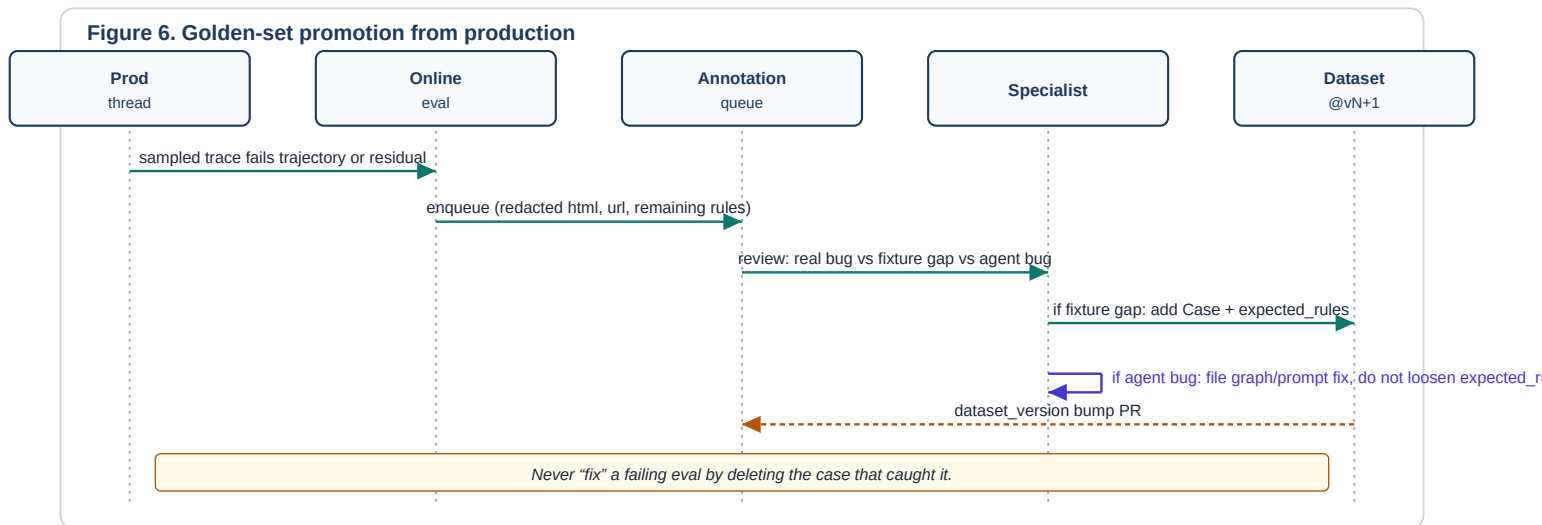
Case schema

```
Case {
  id: "checkout.missing-label.v3"
  url: "https://fixtures.internal/checkout?case=missing-label"
  setup: { viewport: [1280,800], before: [...], cookies?: redacted-ref }
  expected_rules: ["label", "color-contrast"] # must be found on first analyze
  allowed_residual: [] # after a good apply+verify
  tags: ["form", "wcag2aa", "serious"]
  safe_to_auto_apply: false
  notes: "label missing on email field only"
  source: "seed" | "promoted_prod_thread"
  thread_id_src?: "acme/web#7990#..."
}
```

Figure 5. One dataset row. `expected_rules` is ground truth for the first scan.

How the set is built and changed

Source	Rule
Seed fixtures	Checked-in pages (or a fixture app) with planted violations. Start with 15–30 cases covering the rules you actually ship against.
Promotion	On-call adds a failing production thread to an annotation queue. A specialist confirms <code>expected_rules</code> . A new <code>dataset_version</code> is cut.
Versioning	golden-a11y@YYYY.MM.N. CI pins a version. Moving the pin is a reviewed PR, same as a code change.
Retirement	If Axe Platform policy or the product UI makes a case obsolete, mark it retired in that version — do not silently delete history.
Secrets	Fixture pages are public-to-the-eval-runner and contain no real user data. Promoted prod cases must be redacted before they enter the set.
Ownership	Accessibility lead owns <code>expected_rules</code> . Platform team owns the runner. Neither changes the pin casually.



6. How the four planes sit on the graph

intake	[G] allowlist, auth	[O] span intake	
plan	[O] prompt_id on span		
analyze	[G] schema + redact	[O] tool span	[E] seeds outcome
cluster	[G] batch 1..25	[O] batch_sizes	
budget_gate	[G] credits / rounds	[O] credits_used	
remediate	[G] schema + redact	[O] tool span	[E] cost
human_gate	[G] interrupt (IDE)	[O] decision	
apply	[G] file allowlist	[O] files[]	
verify	[G] required edge	[O] remaining_*	[E] outcome judge
report	[G] refuse END if no verify span		
		[D] promotion candidate if residual unexpected	

Figure 7. G = guardrail, O = observability, E = eval hook, D = dataset promotion.

7. SLOs these planes create

SLO	Definition	Plane
Trace completeness	100% of threads have a parent run and a verify span after any apply	Observability
Redaction	0 traces containing fill/cookie/auth values in a weekly sample	Guardrail + Observability
Allowlist	0 analyze calls to hosts outside the list	Guardrail
Eval gate	Candidate graph does not lose outcome score vs baseline on pinned dataset_version	Evals + Dataset
Online drift	Alert if 7-day rate of skipped-verify or budget-overrun rises > 2x	Evals
Dataset freshness	Every serious prod miss is either promoted or explicitly declined within 10 business days	Dataset
Cost	Credits + tokens per fixed serious issue within the agreed band	Observability + Guardrail

8. What this adds to the tool list (still not a shopping spree)

Need	Tool	Already named in Rev 3?
Agent run trees, prompt versions, online/offline evals, annotation queue	LangSmith	Yes — now specified
Infra SLOs, worker health	OpenTelemetry + Prometheus/Grafana	Yes
Deque spend	Axe Account Portal + state.credits_used	Yes
Screenshot / HTML artifacts	Object store + TTL URLs	Yes
Schema and redactor tests	pytest contract suite	Named; now a merge gate

Need	Tool	Already named in Rev 3?
Host and tool allowlists	Graph nodes + adapter, not a new SaaS	Now explicit
Golden cases	LangSmith dataset + fixture app	Named; schema and promotion now explicit
Human labels on fixes	LangSmith annotation queue	Now explicit

No new orchestrator, no second trace backend, no LLM-as-judge for WCAG. If a vendor does not improve one of the four planes in Figure 1, it does not belong in this system.

9. Summary

- **Observe** with a mandatory LangSmith run tree correlated to OTel, credits, and screenshots by thread_id.
- **Guard** with edges and adapters: allowlist, redact, schema, budget, batch size, human/CI graph variants, verify-required.
- **Evaluate** in layers. Axe re-analyze is the outcome judge. Trajectory eval reads the run tree. Online eval samples live traffic.
- **Golden dataset** is versioned, pinned in CI, grown by promoting redacted prod misses, never silently edited to make scores pretty.

Together these four planes are how the Revision 3 stack stops being a demo. The platform diagram from that brief still stands; this brief is the quality architecture that sits on it.